

Course Information for MATH 487 (Dynamical Systems)

Fall 2025

Lectures	Tu Th 4:00–5:15pm in Fine Arts 006
Instructor	Dr. Matthew D. Kvalheim
Email	kvalheim@umbc.edu
Office Hours	Tu Th 2:00–3:00pm in MP 432
Textbook	Differential Dynamical Systems, J. D. Meiss, SIAM, Revised Edition.

1 Course Overview

This course will develop an understanding of fundamental properties of dynamical systems determined by ordinary differential equations. While most differential equations arising in applications do not have closed-form solutions, fundamental questions and qualitative properties—such as existence and uniqueness of solutions, continuous dependence of solutions on initial conditions, and stability—can be investigated for a large class of dynamical systems. This course will focus on the theory behind the rich qualitative behaviors of dynamical systems. Topics include linear time-invariant systems, existence and uniqueness of solutions, stability, (un)stable manifolds, and bifurcations.

We will aim to cover most of chapters 2–5 and part of chapter 8 of the textbook. Specific topics:

- Introduction (chapter 1)
- Linear systems (chapter 2)
 - Linear time-invariant ODE systems, planar systems (§2.1–2.2)
 - Solutions of linear ODEs (§2.3–2.6)
 - Stability of linear ODEs (§2.7)
- Existence and uniqueness of solutions to ODEs (chapter 3)
- Dynamical systems (chapter 4)
 - Flows, equilibria, notions of stability (§4.2, 4.5)

- Linearization and linear stability analysis (§4.4)
- Lyapunov stability, topological conjugacy (§4.6, 4.7)
- Hartman-Grobman theorem, classification of orbits (§4.8–4.10)
- Invariant manifolds (chapter 5)
 - Stable and unstable sets, Hamiltonian systems (§5.1)
 - Stable manifolds (§5.3)
- Bifurcation theory (chapter 8)
 - Bifurcations of equilibria, saddle-node bifurcation (§8.1, 8.4, 8.6)

Please note that these topics are subject to change depending on class progress.

2 Learning Goals and Objectives

- Understand theoretical and computational techniques for solving linear time-invariant ODEs and analyzing stability and other properties;
- Understand existence and uniqueness properties of solutions to nonlinear ODEs;
- Understand how to characterize stability via linearization and Lyapunov functions;
- Understand certain qualitative properties of ODEs, e.g., topological equivalence and asymptotic behavior;
- Understand some basic aspects of bifurcation theory.

3 Homework

Homework will be assigned on an approximate biweekly basis and must be turned in by a specified date. *No late homework will be accepted.* Present your answers neatly and show all of your work; answers without supporting materials may not receive full credit. Homework will be assigned and submitted electronically on Blackboard. You are encouraged to prepare your homework submissions using the mathematical typesetting language L^AT_EX.

4 Exams

There will be 2 in-class midterm exams and 1 final exam. Exam dates will be announced at least 1 week in advance. The final exam is comprehensive. All exams are mandatory and closed book. Calculators and other computing devices are not allowed for any exam.

The final exam is on Tuesday, December 16 at 3:30–5:30pm.

Makeup exams may be given only under **exceptional** circumstances, and only with **written documentation**. Additionally, you must obtain prior approval from the instructor except in case of unexpected medical emergency.

5 Grading Policy

Final grades will be computed using a weighted average of your homework and exam scores with the following weights:

- Homework: 30%
- Exam 1: 20%
- Exam 2: 20%
- Final exam: 30%

If the weighted average is S (between 0 and 100), then grades will be roughly determined as follows:

A	$90 \leq S \leq 100$
B	$80 \leq S < 90$
C	$65 \leq S < 80$
D	$50 \leq S < 65$
F	$S < 50$

However, factors such as the overall distribution of scores may also affect the final letter grade.

6 Academic Integrity

- The UMBC Academic Integrity Statement:

“Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC’s scholarly community in which everyone’s academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct

could result in disciplinary action that may include, but is not limited to, suspension or dismissal. These principles and policies apply in both face-to-face and online classes.”

To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Directory. Resources for students about academic integrity at UMBC are available at <https://academicconduct.umbc.edu/resources-for-students/>.

- All work in homework or an exam must be your own; collaborating on an exam is **not** permitted. Discussions with other students on homework problems are allowed, but your submission must present your own work; simply copying the work of others is a violation of academic integrity.
- If you wish to contest a graded exam, you must make an appeal within one week of the exam return date. All appeals should be made in writing to the instructor with a signed and dated note on the exam. End-of-semester appeals for earlier exams will be ignored.